

The surround cop number of a cubic graph is never two: a refutation of conjecture 00000001223

Submission to The Last Math Competition

Abstract

Conjecture 00000001223 asserts that “the surround cop number of a cubic graph is two”, with tightness of the surround constant attained by the Petersen graph. We show that the statement is false in the strongest possible sense: *no* cubic graph has surround cop number two. In a simple 3-regular graph the robber always stands on a vertex with exactly three distinct neighbours, while two cops occupy at most two vertices; the surround condition — the cops occupy every neighbour of the robber’s vertex — can therefore never hold with two cops, at time zero and hence at no moment of any play, independently of all movement rules. Consequently $\sigma(G) \geq 3$ for every cubic graph G . The smallest witness is K_4 , whose surround cop number is exactly 3; the Petersen graph, Q_3 , $K_{3,3}$ and the triangular prism also all have $\sigma = 3$. Everything is a finite computation: the refutation is formalised in Lean 4 using only the kernel and core, and all seventeen audited theorems depend on no axioms at all.

1 The conjecture

The conjecture reads, verbatim:

Definition: The surround cop number is the least number of cops needed to surround rather than capture. **Conjecture:** The surround cop number of a cubic graph is two; and tightness of the surround constant is attained by the Petersen graph. (*cubic surround constant*)

2 The game and the parameter

The notion at issue is the *surrounding cops and robbers* game of Burgess, Cameron, Clarke, Danziger, Finbow, Jones and Pike [1], who write that in contrast to the original game in which “the cops win by occupying the same vertex as the robber, they now win by occupying each of the robber’s neighbouring vertices”, and that “we denote by $\sigma(G)$ the surrounding cop number of G , namely the least number of cops required to surround a robber in the graph G ”. This matches the definition sentence of the conjecture word for word (“surround rather than capture”) and is the only notion at issue.

Concretely: k cops occupy vertices of G , then the robber occupies a vertex; in each round every cop may stay put or move along an edge (cops never occupy the robber’s vertex), after which the robber must move along an edge to an unoccupied vertex. The cops win as soon as they occupy *every* neighbour of the robber’s current vertex; the robber wins if that never happens. $\sigma(G)$ is the least k for which the cops have a winning strategy.

3 The obstruction: $\sigma(G) \geq \delta(G)$

Lemma 1. *Let G be a simple graph, let v be a vertex of degree d , and let at most $d - 1$ cops be placed on vertices of G (distinct or not). Then the robber standing on v is not surrounded.*

Proof. The d neighbours of v are pairwise distinct, and the at most $d - 1$ cops occupy at most $d - 1$ vertices, so at least one neighbour of v is cop-free. The surround condition — every neighbour of v occupied — therefore fails for this configuration. $\square$

Theorem 1. *For every simple graph G , $\sigma(G) \geq \delta(G)$, where $\delta(G)$ is the minimum degree of G . In particular, for every cubic graph G we have $\sigma(G) \geq 3$, so $\sigma(G) \neq 2$.*

Proof. Suppose $k < \delta(G)$ cops play against G , whatever the movement rules (who moves first, whether players may pass, whether cops move simultaneously or one at a time, when the surround condition is checked). By Lemma 1 the surround condition fails at time zero — before any move has been made, so no rule about moves can matter. Since the cops win only if the condition holds *at some moment*, and it holds at no moment of any play, no cop strategy of any kind wins. Hence k cops do not suffice for every $k < \delta(G)$, i.e. $\sigma(G) \geq \delta(G)$. $\square$

Corollary 1. *The conjecture is false under the universal reading (“every cubic graph has surround cop number two”): by Theorem 1 every cubic graph has $\sigma \geq 3$. It is also false under the existential reading (“some cubic graph has surround cop number two”): no cubic graph attains the value 2 at all. The second conjunct (tightness of the surround constant attained by the Petersen graph) is moot once the first fails; we note that $\sigma(\text{Petersen}) = 3 \neq 2$ as well (Theorem 2 and Table 2).*

4 The witness: $\sigma(K_4) = 3$ exactly

Theorem 2. $\sigma(K_4) = 3$.

Proof. K_4 is simple and cubic. By Theorem 1, $\sigma(K_4) \geq 3$. For the upper bound, place three cops on the three neighbours of the robber’s vertex v (in K_4 these are exactly $V(K_4) \setminus \{v\}$): the surround condition holds at time zero, for every position v of the robber. Hence three cops suffice and $\sigma(K_4) = 3$. $\square$

An exhaustive check strengthens the lower-bound half into a complete verification: over *all* ordered pairs of cop positions in K_4 (including stacked cops, a weaker team than the rules allow) and all robber positions — 64 configurations — the surround condition never holds. The same is done for the other cubic graphs named by the verification pipeline:

graph	n	cubic	2-cop configurations checked	surroundings found
K_4	4	yes	64	0
Petersen	10	yes	1 000	0
Q_3	8	yes	512	0
$K_{3,3}$	6	yes	216	0
triangular prism	6	yes	216	0

Table 1: Exhaustive 2-cop impossibility (all placements, including stacked cops). The surround condition never holds at time zero, hence never at all.

Additionally, a full retrograde (attractor) computation of the game under the standard rules of [1] confirms the exact values:

graph	K_4	Petersen	Q_3	$K_{3,3}$	prism
$\sigma(G)$	3	3	3	3	3

Table 2: Surround cop numbers by game search (`reproduce.py`).

Remark 1 (Robustness of the reading). *The obstruction of Theorem 1 uses only two facts: (i) surround means occupying all neighbours of the robber’s vertex; (ii) cops occupy vertices, at most one cop per vertex (stacking is anyway included in the exhaustive checks and only weakens the cops). Every timing convention in the literature leaves the time-zero argument untouched. Even the strengthened win condition “occupy the closed neighbourhood $N[v]$ ” survives: on K_4 that set has 4 vertices, still more than two cops can occupy; and the robber on a cubic graph always retains a legal move to a cop-free neighbour, so a “robber stuck” loss condition also never triggers with two cops. Finally, K_4 is simple and cubic by any textbook definition, so no reading of “cubic graph” avoids the counterexample.*

5 Formal verification

The Lean 4 formalisation in `lean4/Main.lean` uses only the kernel and core (no Mathlib, no `sorry`). All definitions are pure structural matches over explicit adjacency tables, and every statement is a Boolean equation between closed computable expressions proved by `rfl`, i.e. by kernel reduction alone:

- `surrounded nbr cops v` — the robber on v is surrounded iff every neighbour of v lies in `cops`;
- `twoCopTable nbr n` — exhaustive over every robber vertex $v < n$ and every ordered pair of cop positions (c_1, c_2) , $c_1, c_2 < n$ (stacked cops included): the surround condition never holds;
- `cubicTable nbr n / symmetricTable nbr n` — sanity checks that the explicit tables define simple symmetric cubic graphs;
- `K4_two_cops_never_surround`, `K4_three_cops_surround` — $\sigma(K_4) \geq 3$ and $\sigma(K_4) \leq 3$, i.e. $\sigma(K_4) = 3$;
- `Petersen_two_cops_never_surround`, `Q3_two_cops_never_surround`, `K33_two_cops_never_surround`, `prism_two_cops_never_surround` — the Petersen graph, Q_3 , $K_{3,3}$ and the triangular prism are likewise not 2-surroundable;
- `all_five_two_cops_never_surround` — all five at once.

Running `lake env lean Check.lean` re-evaluates every audited quantity (all come out `true`) and prints `#print axioms` for all seventeen theorems: *every one of them reports “does not depend on any axioms”* — no `propext`, no `Classical.choice`, no `Quot.sound`, no `sorryAx`. The script `reproduce.py` (standard library only) repeats the sanity checks, the exhaustive 2-cop tables, the K_4 witness and the retrograde game search, and exits with status 0 on success.

References

- [1] A. C. Burgess, R. A. Cameron, N. E. Clarke, P. Danziger, S. Finbow, C. W. Jones, D. A. Pike, *Cops that surround a robber*, Discrete Applied Mathematics (2021), <https://doi.org/10.1016/j.dam.2020.06.019>; arXiv:<https://arxiv.org/abs/1910.14200>.